THE COUNT OF CONNECTED COMPONENTS OF GEOMETRIC FIBRES IS WELL DEFINED AND STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7086-7108, immediately following the statement at 7069-7084.

Editorial note. One cross-chapter pointer (Spaces over Fields) is recorded in assembly.external_references.

Editorial note. Same-name pair note: this is the more general side; the schemes side is not counted beside it.

Allowed standard prerequisites (editorial).

- Connected components of a fibre after a separable closure. Spaces over Fields, lemma spaces-over-fields-lemma-separably-closed-field-connected-components, cited at proof lines 7104-7106. Gives bijections between the connected components of the fibres over the various algebraically closed fields representing the same point.
- Equivalence class of a point of an algebraic space. the definition used in the statement (Spec(k) -> Y in the equivalence class of y with k algebraically closed). Fixes the meaning of the function n_{X/Y} so that well-definedness is a statement about that class.
- Base change of the fibre. the morphism g : Y' -> Y in the statement and the resulting fibre X'_{k'} at proof lines 7099-7103. Provides the comparison squares whose vertical maps induce the component bijections.

Source reference. Original reference 0A12: focused-seeds/geometry/S9-spaces-nr-geom-conn-components/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let*

$$n_{X/Y} : |Y| \to \{0, 1, 2, 3, \ldots, \infty\}$$

be the function which associates to $y \in Y$ the number of connected components of X_k where $\mathrm{Spec}(k) \to Y$ is in the equivalence class of y with k algebraically closed. This is well defined and if $g : Y' \to Y$ is a morphism then

$$n_{X'/Y'} = n_{X/Y} \circ g$$

where $X' \to Y'$ is the base change of f.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Suppose that $y' \in Y'$ has image $y \in Y$. Let $\mathrm{Spec}(k') \to Y'$ be in the equivalence class of y' with k' algebraically closed. Then we can choose a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & \mathrm{Spec}(k') & \longrightarrow & Y' \\ & \searrow & & & \downarrow \\ & & \mathrm{Spec}(k) & \longrightarrow & Y \end{array}$$

where K is an algebraically closed field. The result follows as the morphisms of schemes

$$X'_{k'} \longleftarrow (X'_{k'})_K = (X_k)_K \longrightarrow X_k$$

induce bijections between connected components, see Spaces over Fields, Lemma 0A12. To use this to prove the function is well defined take $Y' = Y$. $\square$